Explain the Max-Min method and Max-Product method of fuzzy composition.

2 What is meant by the term defuzzification? Briefly describe the most common method of defuzzification?

3 What is the significance of mutation in a genetic algorithm?

4 Briefly explain the basic foraging behaviour of real ants that is modelled to create the artificial ant colony systems for solving optimization problems.

5 Briefly describe the basic Particle Swarm Optimization (PSO) algorithm.

1 Let U=Animals = {Dog, Cat, Horse, Cow, Lion, Tiger, Elephant} be a universe on which two fuzzy sets, one of Friendly animals and the other one of Strong animals, are defined as shown below.

P=Friendlyanimals={(Dog,0.9),(Cat,1.0),(Horse,0.7),(Cow,0.8),(Lion,0.3),(Tiger, 0.2),(Elephant,0.5)}

Q=Strong animals={(Dog,0.4), (Cat,0.2), (Horse,0.9), (Cow,0.6), (Lion,1.0), (Tiger,1.0), (Elephant,0.8)}

Compute the fuzzy sets $P \cup Q$, $P \cap Q$, P', Q', $P - Q$, $P \oplus Q$ Also verify that $P \cup P' \neq U$ and $P \cap P' \neq \emptyset$

2 Explain the concept of Generalized Modus Tollens (GMT). How does it differ from GMP?

3 Briefly describe multi-objective optimization and pareto optimal set.

4 Explain the process of pheromone updating in Ant Colony Optimization. What are elitist and elastic ants, and how do they contribute to the solution?

5 Discuss the roles and significance of *pbest* and *gbest* particles in solving problems using Particle Swarm Optimization.

Consider the following (discrete) fuzzy sets: A = 0.4/1+0.6/2+0.7/3+0.8/4 and B = 0.3/1+0.65/2+0.4/3+0.1/4

Calculate the fuzzy union A∪ B, the fuzzy intersection A∩ B, and the complement of set A.

What is meant by "defuzzification," and what are the most widely used techniques for defuzzifying in fuzzy logic systems?

Mention the importance of objective (fitness) function in Genetic Algorithm (GA).

Discuss the pros and cons of Ant Colony Optimization (ACO).

What are the key components of Particle Swarm Optimization (PSO) that determine the algorithm's performance?

Consider the set of Colours A= {Blue, Red, Orange, Yellow, Green}, Attributes B = {Bright, Warmth, Dullness}, Feelings C= {Unpleasant, happiness, Angry}. Given R and S where R is the relationship between colours and their attributes and S is the relationship between colour attributes and feelings created. Find the relationship Q between colours and feelings created

R	Bright	Warmth	Dullness
Blue	0.8	0.6	0.4
Red	0.8	0.8	0.2
Orange	0.5	0.7	0.2
Yellow	0.3	0.6	0.5
Green	0.8	0.6	0.4

S	Unpleasant	Happiness	Angry
Bright	0.2	0.8	0.6
Warmth	0.4	0.7	0.8
Dullness	0.8	0.3	0.6

Develop a membership function for “Tall”. Based on that devise membership function for “Very Tall”. Explain how it is done

Mention the importance of objective (fitness) function in genetic algorithm

Describe how pheromone is updated. What is elitist / elastic ants ? Are they useful in this scenario?

What is the significance of pbest and gbest particles in solving problems with particle swarm optimization?